Post-submission candidate: change review

PONE-D-26-50597; internal candidate only. Red strike-through marks removed text; green underline marks added text. Primary frozen outputs are unchanged. This is a review aid, not a revised submission.

~~# Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding~~

Source-only model selection and external evaluation of complete spectral-sharing pipelines in motor-imagery EEG decoding

Corresponding author: zzy2630816871@gmail.com

Postal address: Jinyun Campus, Chongqing Medical University, No. 61, Daxuecheng Middle Road, Shapingba District, Chongqing 401331, China

~~We audited source-only training-duration selection and evaluated fixed shared mu/beta input pipelines for cross-subject motor-imagery EEG decoding. An exploratory four-class study used nine BNCI2014_001 participants, excluding both target sessions from each corresponding model fit and duration selection. Separate frozen binary pipelines were evaluated on PhysioNet (109 people), Cho2017 (52) and Lee2019 offline-training runs (54, two sessions), without target fitting or adaptation. We re-audited saved predictions and source partitions. A BNCI signal analysis, specified after decoder outcomes were known, described baseline-relative mu/beta power and associations with saved binary LOSO balanced accuracy. Mean-rank duration selection increased internal balanced accuracy from 33.72% to 42.67% compared with mean-loss selection; S3 and S8 contributed 87.57% of the aggregate gain. A matched-runtime fixed-duration control improved four of nine people despite a positive mean. Shared-minus-broad external differences were +0.573 percentage points (pp) on PhysioNet (95% person-bootstrap interval [−0.097,1.242]), −1.513 pp on Cho2017 [−2.249,−0.804] and +0.204 pp on Lee2019 [−0.515,0.969]. Cho2017 and Lee2019 Holm-adjusted permutation p-values were 0.000100 and 0.604. These complete-pipeline contrasts used unequal source-selected durations. The BNCI signal analysis retained all 5,184 trials; its 2,592 hand-imagery trials gave mean signed laterality of −0.250 dB in mu and −0.168 dB in beta, with signs varying between people. Training-duration choices yielded large but concentrated gains in this development benchmark. Shared-input pipelines showed no consistent external advantage. BNCI sensor patterns provide descriptive context; nine-person associations do not establish a decoder mechanism. Unknown voltage calibration and offline evaluation limit deployment conclusions.~~

We examined the reliability of source-only training-duration selection and the external performance of complete fixed spectral-sharing pipelines for motor-imagery EEG decoding. An exploratory development study used nine BNCI2014_001 participants, excluding both sessions of each held-out person from fitting and duration selection. Separate frozen binary pipelines were evaluated on PhysioNet, Cho2017 and Lee2019 without target fitting or adaptation. A post-outcome BNCI sensor analysis described baseline-relative mu/beta power without decoder feedback. Mean-rank selection increased development balanced accuracy from 33.72% to 42.67%; S3 and S8 contributed 87.57% of

the aggregate gain. A matched-runtime fixed-duration control improved four of nine people despite a positive mean. Shared-minus-broad external differences were +0.573 percentage points on PhysioNet (95% person-bootstrap interval [−0.097,1.242]), −1.513 on Cho2017 [−2.249,−0.804] and +0.204 on Lee2019 [−0.515,0.969]. These contrasts combine unequal source-selected durations with different input views and computation; they do not isolate spectral sharing. Archived CSP outputs showed strong left-prediction dominance on both later cohorts. The BNCI analysis found negative mean signed hand laterality with participant variation; its nine-person associations do not establish a decoder mechanism. Duration choices yielded large but concentrated development gains. The frozen shared-input pipelines showed no consistent external advantage, and uncertain contrasts establish neither superiority nor equivalence. Acquisition uncertainty and offline evaluation limit deployment conclusions.

~~We address three questions. First, can heterogeneous source-validation loss curves lead to very short training schedules, and does within-fold rank aggregation change those schedules? Second, how does a pipeline that processes predefined mu/beta views with one shared EEGNet compare with broad input, alongside spatial, capacity, and robustness controls? Third, how do the complete source-selected pipelines perform in external binary transfer? A separate descriptive analysis characterizes measured BNCI sensor signals without changing the decoders. The external evidence comprises an earlier PhysioNet evaluation [5,6] and a subsequently frozen, separately fitted pipeline for Cho2017 [7,8] and Lee2019 [9,10]. The evaluations use different channel sets and time windows and are reported separately.~~

The study has two scientific aims. The first is to audit source-only duration selection: how heterogeneous validation-loss curves influence the chosen schedule, and whether an aggregate gain reflects benefit for most participants. The second is to assess complete frozen spectral-sharing pipelines in external binary transfer, retaining adverse and uncertain results alongside developmental spectral, spatial, capacity and robustness controls. The external evidence comprises an earlier PhysioNet evaluation [5,6] and separately fitted, subsequently frozen pipelines for Cho2017 [7,8] and Lee2019 [9,10]. Their montage and time-window differences require separate reporting. A downstream BNCI sensor characterization is auxiliary and receives no role in decoder fitting or selection.

~~Here, we audit source-only model selection as part of cross-subject decoder evaluation [11]. The development analyses trace how training-duration decisions affect apparent model failure and distinguish concentrated aggregate gains from participant-level benefit. Separate frozen external evaluations retain adverse and uncertain outcomes, with participants as the unit of inference. A downstream BNCI sensor analysis adds measured-signal context while keeping physiological description separate from decoder attribution.~~

The methodological contribution is an audit of selection reliability and participant-level heterogeneity, rather than a claim that rank aggregation is a new or generally superior algorithm [11]. The empirical contribution is a frozen external evaluation of complete selected pipelines, including a corrected adverse result and uncertain estimates. Saved predictions, contracts and checksums make these claims inspectable; that evidence chain is a transparency contribution, not a new decoding method. The physiological characterization supplies descriptive sensor context rather than mechanistic validation.

Multibranch designs are relevant architectural precedents, but they do not define the same evaluation. Roots et al. combine three separately parameterized EEGNet-style branches [23]; their

published protocol randomly splits pooled half-second windows, rather than documenting an unseen-person external test. Luo et al. use independently learned branches for overlapping frequency bands [24], primarily in within-dataset multisubject/session evaluation. Neither design is equivalent to fixed mu/beta inputs passed through one shared backbone.

Training stability and transfer are also studied outside this information contract. Smedemark-Margulies et al. examine divergence regularization in subject-disjoint rapid-serial-visual-presentation EEG [25]; this is not a motor-imagery benchmark or a direct test of the present duration-selection rules. Zhang et al. describe target-subject fine-tuning of pretrained convolutional networks [26]. The motor-imagery task of the BEETL competition provides labelled target calibration [27], unlike the present frozen no-target-fitting transfer.

Recent PLOS ONE models illustrate further protocol differences. SMMTM uses learned multiscale temporal branches and reports within-dataset leave-one-subject-out evaluation [28]. Its validation-selection details do not establish a matching frozen external budget. RMETNet incorporates source–target distribution matching and Riemannian alignment [29], so it cannot serve as a source-only comparator without separating those operations. The protocol audit in S1 Appendix explains these distinctions without inferring unreported leakage or ranking headline scores.

Sonntag et al. relate resting spectral features to participant-specific unilateral upper-limb motor-imagery classifiers in a 26-person study [30]. Their task, resting-power estimator and within-person evaluation differ from the present nine-person post-outcome, baseline-relative sensor descriptors. That literature motivates physiological description; it does not independently validate the present associations or identify a learned decoder mechanism.

~~The study comprises exploratory development and two external-evaluation stages. In the four-class BNCI2014_001 development stage, both sessions of each target person are excluded from the corresponding fitting and duration selection. Earlier benchmark outcomes informed later candidates, so the nine-person comparisons remain exploratory. The earlier PhysioNet external evaluation (Q14) applies separately fitted binary BNCI models to PhysioNet; a metadata-driven sampling-rate amendment followed partial target execution. The frozen Cho2017/Lee2019 transfer experiment (Q15) uses a new binary source pipeline with a different montage, crop and reference, evaluating its frozen broad, shared-input and CSP models on Cho2017 and Lee2019 offline-training segments. These datasets are external to fitting, but participant identity nonoverlap across providers was not independently established.~~

The study comprises exploratory development and two external-evaluation stages. In the four-class BNCI2014_001 development stage, both sessions of each target person are excluded from the corresponding fitting and duration selection. Earlier benchmark outcomes informed later candidates, so the nine-person comparisons remain exploratory. The earlier PhysioNet external evaluation applies separately fitted binary BNCI models to PhysioNet; a metadata-driven sampling-rate amendment followed partial target execution. The frozen Cho2017/Lee2019 transfer experiment uses a new binary source pipeline with a different montage, crop and reference, evaluating its frozen broad, shared-input and CSP models on Cho2017 and Lee2019 offline-training segments. These datasets are external to fitting, but participant identity nonoverlap across providers was not independently established.

~~The primary external estimand is the mean paired balanced-accuracy difference between the frozen shared-input and broad pipelines, with equal weight for each person and separate estimates for each~~

~~cohort. The PhysioNet evaluation uses 17 shared and 18 broad epochs; the frozen Cho2017/Lee2019 evaluation uses 19 shared and 14 broad epochs, all selected on source data. Each contrast therefore evaluates complete selected pipelines rather than the isolated effect of sharing. CSP provides context, and neither external stage compares rank with mean-loss selection. The descriptive BNCI physiological analysis (Q16) was defined after decoder results were known and uses BNCI signals only. Its raw-signal summaries and associations provide no feedback into the decoders and cannot establish their physiological mechanism.~~

The primary external estimand is the mean paired balanced-accuracy difference between the frozen shared-input and broad pipelines, with equal weight for each person and separate estimates for each cohort. The PhysioNet evaluation uses 17 shared and 18 broad epochs; the frozen Cho2017/Lee2019 evaluation uses 19 shared and 14 broad epochs, all selected on source data. Each contrast therefore evaluates complete selected pipelines rather than the isolated effect of sharing. CSP provides context, and neither external stage compares rank with mean-loss selection. The descriptive BNCI physiological analysis was defined after decoder results were known and uses BNCI signals only. Its raw-signal summaries and associations provide no feedback into the decoders and cannot establish their physiological mechanism.

~~The corresponding saved summary is presented in Fig 1.~~

Fig 1 summarizes the study sequence and source–target information boundaries.

~~Fig 1. Study workflow and source–target information boundaries. Study workflow and information boundaries. Both sessions of each internal target person are excluded from the corresponding source fit; these development analyses remain exploratory. Q14 and Q15 use separate binary models and preprocessing contracts. At inference, deterministic adapters and frozen source-derived transformations are applied without fitting, adaptation or BatchNorm updates. Event and class metadata establish eligibility and canonical label mapping; ground-truth labels are used for scoring. The separate BNCI-only physiological analysis reads native signals and labels after decoder outcomes are known, without feeding back into fitting or selection. Results are aggregated by person. Q14's post-partial-run amendment and the external differences in selected duration are described in the Methods and comparison-fairness table.~~

Fig 1. Study workflow and information boundaries. Both sessions of each internal target person are excluded from the corresponding source fit; these development analyses remain exploratory. The PhysioNet and Cho2017/Lee2019 evaluations use separate binary models and preprocessing contracts. At inference, deterministic adapters and frozen source-derived transformations are applied without fitting, adaptation or BatchNorm updates. Event and class metadata establish eligibility and canonical label mapping; ground-truth labels are used for scoring. The separate BNCI-only physiological analysis reads native signals and labels after decoder outcomes are known, without feeding back into fitting or selection. Results are aggregated by person. The earlier PhysioNet evaluation's post-partial-run amendment and the external differences in selected duration are described in the Methods and comparison-fairness table.

~~The corresponding saved summary is presented in Table 1.~~

Table 1 distinguishes source and external scoring populations.

~~| Q14 binary BNCI source | 9 | 2 | 2,592 | 22 × 480 at 160 Hz |~~

| PhysioNet-stage binary BNCI source | 9 | 2 | 2,592 | 22 × 480 at 160 Hz |

~~| Q14 external PhysioNet | 109 | 2 | 4,918 | 22 × 480 at 160 Hz |~~

| External PhysioNet | 109 | 2 | 4,918 | 22 × 480 at 160 Hz |

~~| Q15 binary BNCI source | 9 | 2 | 2,592 | 21 × 320 at 160 Hz |~~

| Cho/Lee-stage binary BNCI source | 9 | 2 | 2,592 | 21 × 320 at 160 Hz |

~~| Q15 external Cho2017 | 52 | 2 | 10,520 | 21 × 320 at 160 Hz |~~

| External Cho2017 | 52 | 2 | 10,520 | 21 × 320 at 160 Hz |

~~| Q15 external Lee2019 | 54 | 2 | 10,800 | 21 × 320 at 160 Hz |~~

| External Lee2019 | 54 | 2 | 10,800 | 21 × 320 at 160 Hz |

~~Both sessions of each BNCI outer target are held out. Q14 and Q15 have separate binary source fits and preprocessing contracts, and neural inputs contain EEG channels only. Prediction rows repeat trials across models and seeds; they are not independent participants. Lee sessions are pooled within person for the primary endpoint.~~

Both sessions of each BNCI outer target are held out. The PhysioNet and Cho2017/Lee2019 evaluations have separate binary source fits and preprocessing contracts, and neural inputs contain EEG channels only. Prediction rows repeat trials across models and seeds; they are not independent participants. Lee sessions are pooled within person for the primary endpoint.

~~Target EEG provides model inputs; event and class metadata determine declared eligibility and canonical label mappings. Labels are then used to score saved predictions. Cho recordings are stored in class-specific containers, so the adapter has access to native class metadata. Fixed filtering, channel selection, resampling and common-average referencing operate on target signals without fitting a target-dependent decoder. The Q15 Helmert basis depends only on channel count; scalers, CSP/PCA, whitening and EOG-regression coefficients, where used, are estimated on source-training data and applied unchanged. Separate physiological summaries use raw signals and labels but provide no feedback into bands, crops, weights, thresholds, schedules or model choice.~~

Target EEG provides model inputs; event and class metadata determine declared eligibility and canonical label mappings. Labels are then used to score saved predictions. Cho recordings are stored in class-specific containers, so the adapter has access to native class metadata. Fixed filtering, channel selection, resampling and common-average referencing operate on target signals without fitting a target-dependent decoder. The Cho/Lee fixed Helmert basis depends only on channel count; scalers, CSP/PCA, whitening and EOG-regression coefficients, where used, are estimated on source-training data and applied unchanged. Separate physiological summaries use raw signals and labels but provide no feedback into bands, crops, weights, thresholds, schedules or model choice.

~~The absence of target fitting does not establish physical acquisition calibration. Q15's native-numeric-as-microvolt convention does not verify the exported voltage units. Earlier within-person and cross-session P2 models use target-person labelled calibration and remain outside the strict zero-calibration endpoints. The source-fitted P4 EOG regression requires synchronous target EOG and thus additional sensors, although it fits no target coefficients.~~

The absence of target fitting does not establish physical acquisition calibration. The Cho2017/Lee2019 evaluation's native-numeric-as-microvolt convention does not verify the exported voltage units. Earlier within-person and cross-session calibrated models use target-person labelled calibration and remain outside the strict zero-calibration endpoints. The source-fitted EOG regression requires synchronous target EOG and thus additional sensors, although it fits no target coefficients.

~~The corresponding saved summary is presented in Table 2.~~

Table 2 specifies the target-information budget.

~~| Filtering, resampling, crop and Q15 CAR | Target EEG, fixed native metadata | None | Deterministic offline operations |~~

| Filtering, resampling, crop and Cho/Lee common-average reference | Target EEG, fixed native metadata | None | Deterministic offline operations |

~~| Q15 Helmert coordinates | Channel count and signal | None | Analytical CAR-subspace basis |~~

| Cho/Lee fixed Helmert coordinates | Channel count and signal | None | Analytical CAR-subspace basis |

~~| Scoring and separate Q16 characterization | Predictions/raw signals and canonical labels | None | Downstream analyses with no decoder feedback |~~

| Scoring and separate BNCI physiology characterization | Predictions/raw signals and canonical labels | None | Downstream analyses with no decoder feedback |

~~| P2 within-person or cross-session models | Target-person labeled training trials | Yes | Calibrated context; separate endpoint |~~

| Within-person or cross-session calibrated models | Target-person labeled training trials | Yes | Calibrated context; separate endpoint |

~~The mean-loss rule selects the earliest epoch that minimizes validation cross-entropy averaged with equal weight across the four inner folds. Candidate epochs are 1–40. The mean-rank rule ranks epochs within each fold by validation cross-entropy, assigns average ranks to ties, averages the ranks across folds, and selects the earliest minimum. This rule is invariant to strictly increasing (order-preserving) transformations of the losses within each fold, although that property does not guarantee better target prediction. The rank follow-up reuses the 36 frozen original inner trajectories and trains 27 new final models after the schedule has been frozen.~~

The mean-loss rule selects the earliest epoch that minimizes validation cross-entropy averaged with equal weight across the four inner folds. Candidate epochs are 1–40. The mean-rank rule ranks epochs within each fold by validation cross-entropy, assigns average ranks to ties, averages the ranks across folds, and selects the earliest minimum. This rule is invariant to strictly increasing (order-preserving) transformations of the losses within each fold, although that property does not guarantee better target prediction. The rank follow-up reuses the 36 frozen original inner trajectories and trains 27 new final models after the schedule has been frozen. Equation (1) states the selected-epoch rule.

~~Later controls compare fixed 20-epoch training with raw-loss-selected schedules. A matched-runtime experiment retains the historical source-selected epoch counts but retrains the 27 final models using the same runtime and raw files as the fixed-duration comparator. Epochs are not reselected under the newer runtime. Source-count controls use k = 2, 4, or 6 source participants, four predefined identity subsets per count, and the k = 8 anchor; all counts use fixed 20 epochs. The 0train-versus-1test source-session contrast has equal training budgets: 2,304 trials, 20 epochs, and 720 optimizer updates per fit. Source-count contrasts and comparisons of one source session with the both-session anchor change both the number of training examples and optimizer updates, preventing attribution to a single causal factor.~~

Later controls compare fixed 20-epoch training with raw-loss-selected schedules. A matched-runtime experiment retains the historical source-selected epoch counts but retrains the 27 final models using the same runtime and raw files as the fixed-duration comparator. Epochs are not reselected under the newer runtime. Source-count controls use k = 2, 4, or 6 source participants, four predefined identity subsets per count, and the k = 8 anchor; all counts use fixed 20 epochs. The 0train-versus-1test source-session contrast has equal training budgets: 2,304 trials, 20 epochs, and 720 optimizer updates per fit. Source-count contrasts and comparisons of one source session with the both-session anchor change both the number of training examples and optimizer updates, preventing attribution to a single causal factor. Twenty epochs was a fixed development follow-up budget, not a prospectively established optimum or a recommendation for external deployment.

~~The population raw-versus-source-normalized Q5/Q6 comparison also changes the selected duration for S3 from 2 to 16 epochs and S5 from 13 to 12. Its aggregate difference therefore cannot be attributed solely to normalization. The fixed-duration Q7 four-cell diagnostic separates normalization and duration more directly, but only for S3.~~

The population raw-versus-source-normalized comparison also changes the selected duration for S3 from 2 to 16 epochs and S5 from 13 to 12. Its aggregate difference therefore cannot be attributed solely to normalization. The fixed-duration four-cell diagnostic separates normalization and duration more directly, but only for S3.

~~For shared mu/beta input, both band views pass through the same EEGNet in one concatenated mini-batch. Their class logits are averaged before cross-entropy during training and before softmax during inference. Network parameters and BatchNorm are shared, while dropout acts per view. The model has no learned band weights, attention mechanism, or independently trained branches. Its parameter count matches one EEGNet, but forward-pass computation and preprocessing differ.~~

For shared mu/beta input, both band views pass through the same EEGNet in one concatenated mini-batch. Their class logits are averaged before cross-entropy during training and before softmax during inference. Network parameters and BatchNorm are shared, while dropout acts per view. The model has no learned band weights, attention mechanism, or independently trained branches. Its parameter count matches one EEGNet, but forward-pass computation and preprocessing differ. Equation (2) defines the shared-view prediction.

~~The corresponding saved summary is presented in Table 3.~~

Source-only Robustness Interventions

The six completed robustness conditions are source-pooled whitening, source-balanced empirical risk minimization (ERM), source-group distributionally robust optimization (GroupDRO), channel

dropout, gain perturbation, and their combination. Source whitening uses equal-person covariance aggregation, a ridge of 5% of the mean eigenvalue, and scale preservation; no target covariance or alignment is fitted. GroupDRO treats each source person as a group [31]. Persistent log weights increase by 0.05 times detached per-group cross-entropy, are normalized by log-sum-exp, and weight the group loss. Its matched comparator is balanced ERM, which preserves the batching/sampling intervention. Training-only channel dropout uses probability 0.1 with surviving-channel scale 1/0.9; gain perturbation uses U[0.8,1.2] and precedes dropout in the combined condition. Validation and target inputs are not augmented.

Whitening and augmentation conditions are compared with the previously frozen broad rank-selected baseline. The historical baseline ran on a different runtime from the robustness batch, so these differences reflect both intervention and runtime variation. GroupDRO versus balanced ERM provides a within-runtime comparison of training objectives. This distinction accompanies the numerical table and limits attribution of cross-batch differences to the named interventions.

Earlier Binary Source Freeze and PhysioNet Evaluation

The binary BNCI source population contains 2,592 left/right trials, 288 per person. Filtering and epoching occur at 250 Hz before deterministic polyphase resampling by 16/25 to 160 Hz, giving 22 × 480 samples. The broad binary model uses 8–30 Hz; the shared model uses the fixed mu/beta views. Source-only duration selection across all nine participants uses validation groups comprising people 1–2, 3–4, 5–6, and 7–9, with equal fold weighting. The frozen training durations are 18 epochs for the broad model and 17 for the shared model. Checkpoints, channels, labels, contrast, seeds, and preprocessing are fixed before external access. The source-only CSP4 + LDA comparator uses reg = None and LDA solver = svd; it differs from the internal CSP8 OAS/shrinkage-LDA configuration. The external neural comparison evaluates complete pipelines with different source-selected durations and therefore cannot isolate parameter sharing.

A partial external run stopped at S88 because its 128-Hz header violated the original 160-Hz assumption. A versioned amendment used metadata from the remaining headers without consulting predictions or scores. S88, S92, and S100 are filtered and epoched at their native rate, then resampled by 5/4 from 384 to 480 samples. The other 106 subjects retain native 160-Hz processing. The 87 completed subject outputs from the earlier run are copied byte-for-byte, and 22 additional subjects complete the cohort. The amendment introduces no target statistics, fitted normalization, adaptation, seed selection, or class-performance-dependent processing. Because it followed partial external execution, the combined result does not represent an unmodified execution of the original freeze.

The original aggregate validation failed because it applied an inappropriate exact floating-point equality check to re-serialized CSV probabilities. A separate cross-host runtime mismatch later prompted a versioned portability amendment. Both limitations are documented. The versioned validation-only audit checked official EDF checksums and event identities and recomputed saved probability/argmax consistency, subject metrics, and statistics. It passed with zero new fits and zero new model-inference rows. It therefore validates saved outputs; it does not regenerate all external predictions from checkpoints on the later validation host.

Frozen Cho2017/Lee2019 Transfer

The frozen Cho2017/Lee2019 study is a separate binary transfer experiment using different checkpoints from the earlier PhysioNet evaluation. All nine BNCI participants supply 2,592 left/right

trials. The common ordered montage comprises Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, and POz. Each trial is processed in a fixed native-rate context from −1.5 to +4.5 s relative to the cue. Common-average reference is computed across these 21 selected channels, followed by native-rate zero-phase second-order-section Butterworth filtering with prototype order four. Broad input uses 8–30 Hz; the shared views use 8–13 and 13–30 Hz. Deterministic polyphase resampling yields 160 Hz, followed by the half-open cue crop [0.5,2.5) s, equivalent to samples [320,640) of the resampled six-second context. The resulting model tensor is 21 × 320. These preprocessing and input differences preclude interpreting a comparison with the earlier PhysioNet evaluation as an effect of external provider alone.

The Cho2017/Lee2019 transfer adapter interprets native numeric values as microvolts by convention. Physical voltage calibration, the original Cho hardware reference, and hardware cue latency have not been independently verified. The first 64 of the Cho payload's 68 rows are EEG; four EMG rows are excluded. Channel mapping follows the publication's numbered montage because the payload does not supply names. Cho native labels left = 1 and right = 2 are retained. Lee native labels left = 2 and right = 1 are deterministically mapped to canonical left = 1 and right = 2. Lee MATLAB cue indices are converted using t − 1; the provider's stored segments align with Python t, leaving a disclosed one-sample (1-ms) indexing discrepancy that was not changed using outcome scores. The supplement documents these adapter assumptions and the file-level checks.

Source-only duration selection uses four groups comprising people 1–2, 3–4, 5–6, and 7–9, candidate epochs 1–40, equally weighted mean within-fold cross-entropy ranks, and the earliest minimum. Final durations are 14 broad epochs and 19 shared epochs. The selection seed is 20260923, and final seeds are 20260924–20260926. Training uses the same configured EEGNet family, optimizer, and fixed-band logit averaging described above, with the 21-channel, 320-sample transfer dimensions. A source-fitted CSP4 + LDA comparator applies a fixed 21-to-20 Helmert coordinate transform after common-average reference, followed by source-only fitting. The external neural comparison evaluates two complete frozen pipelines with different source-selected durations and cannot isolate parameter sharing or band decomposition alone. Source training comprises eight inner neural fits, six final neural fits, and one shallow fit: 15 in total. Migration reused these artifacts without additional source or target fitting. The broad model processes one view and the shared model two views; normalization exposure and computation also differ. These differences accompany the unequal durations in the complete-pipeline contrast.

Table 3 records matched aspects and remaining differences.

~~| Q5 mean-loss vs Q8 mean-rank | Nine-person all-trial LOSO, 22×750, 4–40 Hz, original 36 inner trajectories, final seeds | Schedule changes; candidate arose after BNCI outcomes; exploratory |~~

| Mean-loss versus mean-rank duration selection | Nine-person all-trial LOSO, 22×750, 4–40 Hz, original 36 inner trajectories, final seeds | Schedule changes; candidate arose after BNCI outcomes; exploratory |

~~| Q5 raw vs Q6 source normalization | Same all-trial targets and EEGNet family; source-only scaler fit | Selected epochs also change: S3 2→16, S5 13→12; not normalization alone |~~

| Raw versus source-normalized input | Same all-trial targets and EEGNet family; source-only scaler fit | Selected epochs also change: S3 2→16, S5 13→12; not normalization alone |

~~| Q7 S3 raw/normalized ×2/16 epochs | Epochs fixed within each raw/normalized pair | One-person diagnostic |~~

| S3 raw/normalized ×2/16 epochs | Epochs fixed within each raw/normalized pair | One-person diagnostic |

~~| Q9 broad vs shared, reselected durations | Same targets, dimensions, EEGNet parameter count, optimizer/seeds | Bands, two-view computation, BatchNorm exposure and schedules differ |~~

| Broad versus shared, reselected durations | Same targets, dimensions, EEGNet parameter count, optimizer/seeds | Bands, two-view computation, BatchNorm exposure and schedules differ |

~~| Q9-E002 fixed-schedule broad/shared | Same targets and frozen Q8 per-target epochs | Duration-controlled sensitivity; views/computation still differ |~~

| Broad/shared, fixed per-target schedule | Same targets and frozen mean-rank per-target epochs | Duration-controlled sensitivity; views/computation still differ |

~~| Q10 CSP vs PCA neural projections | Eight source-fitted components per band; 16 projected channels and source scaling | Matched to each other, not to 22-channel shared-logit input |~~

| CSP versus PCA neural projections | Eight source-fitted components per band; 16 projected channels and source scaling | Matched to each other, not to 22-channel shared-logit input |

~~| Q11 independent branches vs capacity comparator | Same targets; approximate trainable capacity | 5,864 vs 5,914 parameters; branch computation/normalization differ |~~

| Independent branches versus capacity comparator | Same targets; approximate trainable capacity | 5,864 vs 5,914 parameters; branch computation/normalization differ |

~~| Q12 GroupDRO vs balanced ERM | Same runtime and batching/sampling policy | Declared objective comparison |~~

| GroupDRO versus balanced ERM | Same runtime and batching/sampling policy | Declared objective comparison |

~~| Q12 whitening/augmentation vs historical Q8 | Same nominal target population and source-only rule | Historical baseline uses another runtime |~~

| Whitening/augmentation versus historical mean-rank | Same nominal target population and source-only rule | Historical baseline uses another runtime |

~~| Q13 fixed20 vs matched-runtime CE schedule | Same actual runtime, raw files and three seeds | Historical epochs replayed, not reselected in the newer runtime |~~

| Fixed20 versus matched-runtime CE schedule | Same actual runtime, raw files and three seeds | Historical epochs replayed, not reselected in the newer runtime |

~~| Q13 source-session 0train vs 1test | 2,304 source trials, 20 epochs, 720 updates | Session-composition changes; both-session anchor also changes exposure |~~

| Source-session 0train versus 1test | 2,304 source trials, 20 epochs, 720 updates | Session composition changes; both-session anchor also changes exposure |

~~| Q14 external broad vs shared | 109 people/4,918 trials; 22×480; same source seeds and target rule | 18 vs 17 epochs; metadata amendment after partial execution |~~

| PhysioNet broad versus shared | 109 people/4,918 trials; 22×480; same source seeds and target rule | 18 vs 17 epochs; metadata amendment after partial execution |

~~| Q15 external broad vs shared | Identical people/trials within cohort; 21×320 adapter, neural family/seeds | 14 vs 19 epochs and one-view vs two-view processing |~~

| Cho2017/Lee2019 broad versus shared | Identical people/trials within cohort; 21×320 adapter, neural family/seeds | 14 vs 19 epochs and one-view vs two-view processing |

~~| Q14/Q15 CSP4+LDA context | Same cohort scoring population; source-only fit | One shallow model vs three neural seeds; Q15 Helmert; differs from Q4 CSP8/OAS |~~

| External CSP4+LDA context | Same cohort scoring population; source-only fit | One shallow model vs three neural seeds; Cho/Lee fixed Helmert; differs from traditional-baseline CSP8/OAS |

~~Each comparison controls the aspects listed, while retaining the stated implementation differences. Q14/Q15 therefore compare complete frozen pipelines. Matching selected aspects does not isolate every causal factor, and the CSP comparisons remain descriptive.~~

Each comparison controls the aspects listed, while retaining the stated implementation differences. The two external evaluations therefore compare complete frozen pipelines. Matching selected aspects does not isolate every causal factor, and the CSP comparisons remain descriptive.

~~### Source-only Robustness Interventions~~

~~The six completed robustness conditions are source-pooled whitening, source-balanced empirical risk minimization (ERM), source-group distributionally robust optimization (GroupDRO), channel dropout, gain perturbation, and their combination. Source whitening uses equal-person covariance aggregation, a ridge of 5% of the mean eigenvalue, and scale preservation; no target covariance or alignment is fitted. GroupDRO treats each source person as a group [23]. Persistent log weights increase by 0.05 times detached per-group cross-entropy, are normalized by log-sum-exp, and weight the group loss. Its matched comparator is balanced ERM, which preserves the batching/sampling intervention. Training-only channel dropout uses probability 0.1 with surviving channel scale 1/0.9; gain perturbation uses U[0.8,1.2] and precedes dropout in the combined condition. Validation and target inputs are not augmented.~~

~~Whitening and augmentation conditions are compared with the previously frozen broad rank-selected baseline. The historical baseline ran on a different runtime from the robustness batch, so these differences reflect both intervention and runtime variation. GroupDRO versus balanced ERM provides a within-runtime comparison of training objectives. This distinction accompanies the numerical table and limits attribution of cross-batch differences to the named interventions.~~

~~### Earlier Binary Source Freeze and PhysioNet Evaluation (Q14)~~

The binary BNCI source population contains 2,592 left/right trials, 288 per person. Filtering and epoching occur at 250 Hz before deterministic polyphase resampling by 16/25 to 160 Hz, giving 22 × 480 samples. The broad binary model uses 8–30 Hz; the shared model uses the fixed mu/beta views. Source-only duration selection across all nine participants uses validation groups comprising people 1–2, 3–4, 5–6, and 7–9, with equal fold weighting. The frozen training durations are 18 epochs for the broad model and 17 for the shared model. Checkpoints, channels, labels, contrast, seeds, and preprocessing are fixed before external access. The source-only CSP4 + LDA comparator uses reg = None and LDA solver = svd; it differs from the internal CSP8 OAS/shrinkage-LDA configuration. The external neural comparison evaluates complete pipelines with different source-selected durations and therefore cannot isolate parameter sharing.

A partial external run stopped at S88 because its 128-Hz header violated the original 160-Hz assumption. A versioned amendment used metadata from the remaining headers without consulting predictions or scores. S88, S92, and S100 are filtered and epoched at their native rate, then resampled by 5/4 from 384 to 480 samples. The other 106 subjects retain native 160-Hz processing. The 87 completed subject outputs from the earlier run are copied byte-for-byte, and 22 additional subjects complete the cohort. The amendment introduces no target statistics, fitted normalization, adaptation, seed selection, or class-performance-dependent processing. Because it followed partial external execution, the combined result does not represent an unmodified execution of the original freeze.

The original aggregate validation failed because it applied an inappropriate exact floating-point equality check to re-serialized CSV probabilities. A separate cross-host runtime mismatch later prompted a versioned portability amendment. Both limitations are documented. The versioned validation-only audit checked official EDF checksums and event identities and recomputed saved probability/argmax consistency, subject metrics, and statistics. It passed with zero new fits and zero new model-inference rows. It therefore validates saved outputs; it does not regenerate all external predictions from checkpoints on the later validation host.

Frozen Cho2017/Lee2019 Transfer (Q15)

The frozen Cho2017/Lee2019 study is a separate binary transfer experiment using different checkpoints from Q14. All nine BNCI participants supply 2,592 left/right trials. The common ordered montage comprises Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, and POz. Each trial is processed in a fixed native-rate context from −1.5 to +4.5 s relative to the cue. Common-average reference is computed across these 21 selected channels, followed by native-rate zero-phase second-order-section Butterworth filtering with prototype order four. Broad input uses 8–30 Hz; the shared views use 8–13 and 13–30 Hz. Deterministic polyphase resampling yields 160 Hz, followed by the half-open cue crop [0.5,2.5) s, equivalent to samples [320,640) of the resampled six-second context. The resulting model tensor is 21 × 320. These preprocessing and input differences preclude interpreting a comparison with Q14 as an effect of external provider alone.

The Q15 adapter interprets native numeric values as microvolts by convention. Physical voltage calibration, the original Cho hardware reference, and hardware cue latency have not been independently verified. The first 64 of the Cho payload's 68 rows are EEG; four EMG rows are excluded. Channel mapping follows the publication's numbered montage because the payload does not supply names. Cho native labels left = 1 and right = 2 are retained. Lee native labels left = 2 and right = 1 are deterministically mapped to canonical left = 1 and right = 2. Lee MATLAB cue indices are converted using t − 1; the provider's stored segments align with Python t, leaving a disclosed one-

~~sample (1-ms) indexing discrepancy that was not changed using outcome scores. The supplement documents these adapter assumptions and the file-level checks.~~

~~Source-only duration selection uses four groups comprising people 1–2, 3–4, 5–6, and 7–9, candidate epochs 1–40, equally weighted mean within-fold cross-entropy ranks, and the earliest minimum. Final durations are 14 broad epochs and 19 shared epochs. The selection seed is 20260923, and final seeds are 20260924–20260926. Training uses the same configured EEGNet family, optimizer, and fixed-band logit averaging described above, with the Q15 channel/time dimensions. A source-fitted CSP4 + LDA comparator applies a fixed 21-to-20 Helmert coordinate transform after common-average reference, followed by source-only fitting. The external neural comparison evaluates two complete frozen pipelines with different source-selected durations and cannot isolate parameter sharing or band decomposition alone. Source training comprises eight inner neural fits, six final neural fits, and one shallow fit: 15 in total. Migration reused these artifacts without additional source or target fitting.~~

~~### BNCI-only Physiological Signal Characterization (Q16)~~

BNCI-only Physiological Signal Characterization

~~The physiological parameters were fixed on 6 October 2026 after the decoder outcomes were known and before the physiological calculations. The analysis is therefore descriptive and does not provide independent confirmation. Associations use existing matched-binary Q14-E001 BNCI LOSO scores, averaged over the three fixed neural seeds within each of the same nine people; all-nine-source fits and external-cohort scores are excluded. The broad EEGNet score is the designated descriptive endpoint. Six Spearman coefficients cover the three fixed models and two fixed bands, with no p-values, tertile groups or outcome-selected bands/models. Decoder imagery covers [0.5,3.5) s, whereas the physiological power analysis covers [0.5,2.5) s. No external physiology was calculated, and no physiological output was used to fit, select or update a decoder. The parameter freeze is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a; its public readback preceded raw-power calculation. The preprocessing freeze, raw-data receipt, execution manifest and final independent validation link the specification and inputs to the completed tables in Q16-P001-BNCI-20261006.~~

The physiological parameters were fixed on 6 October 2026 after the decoder outcomes were known and before the physiological calculations. The analysis is therefore descriptive and does not provide independent confirmation. Associations use existing matched-binary BNCI LOSO scores, averaged over the three fixed neural seeds within each of the same nine people; all-nine-source fits and external-cohort scores are excluded. The broad EEGNet score is the designated descriptive endpoint. Six Spearman coefficients cover the three fixed models and two fixed bands, with no p-values, tertile groups or outcome-selected bands/models. Decoder imagery covers [0.5,3.5) s, whereas the physiological power analysis covers [0.5,2.5) s. No external physiology was calculated, and no physiological output was used to fit, select or update a decoder. The parameter freeze is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a; its public readback preceded raw-power calculation. The preprocessing freeze, raw-data receipt, execution manifest and final independent validation link the specification and inputs to the completed tables identified in the supplementary provenance inventory.

~~The held-out person is the unit of statistical inference. Paired seeds are averaged within person; predefined source-identity subsets are averaged within person and source count. Shallow conditions have one prediction per trial rather than three independent neural seeds. Paired differences are~~

~~expressed in balanced-accuracy percentage points (pp). Person-bootstrap percentile intervals resample people, not trials, sessions, seeds, or source subsets. External intervals describe variation across target people conditional on the frozen source models; they do not quantify source-cohort sampling uncertainty. Internal analyses remain exploratory because benchmark outcomes informed the research program and outer training sets overlap. For Lee2019, the two retained sessions are pooled before calculating each person/seed balanced accuracy. The three neural-seed scores are then averaged within person, giving an average of scores rather than an ensemble of probabilities. A single CSP score is retained per person. Cross-cohort score differences are descriptive because Q14 and Q15 differ in source preprocessing and checkpoints.~~

The held-out person is the unit of statistical inference. Paired seeds are averaged within person; predefined source-identity subsets are averaged within person and source count. Shallow conditions have one prediction per trial rather than three independent neural seeds. Paired differences are expressed in balanced-accuracy percentage points (pp). Person-bootstrap percentile intervals resample people, not trials, sessions, seeds, or source subsets. External intervals describe variation across target people conditional on the frozen source models; they do not quantify source-cohort sampling uncertainty. Internal analyses remain exploratory because benchmark outcomes informed the research program and outer training sets overlap. For Lee2019, the two retained sessions are pooled before calculating each person/seed balanced accuracy. The three neural-seed scores are then averaged within person, giving an average of scores rather than an ensemble of probabilities. A single CSP score is retained per person. Cross-cohort score differences are descriptive because the PhysioNet and Cho2017/Lee2019 evaluations differ in source preprocessing and checkpoints. Equation (3) defines the balanced-accuracy scoring rule.

~~For each archived experiment, we report its original statistics and the interval resampling procedure. Rank-selection intervals use 20,000 person-bootstrap draws with seed 20260923. The archived shared-minus-broad representation contrast and robustness intervals use 20,000 draws with seed 20260924. Source-composition and matched-runtime intervals use 10,000 draws with seed 20260926 and exhaustive two-sided sign flips over nine participant differences; familywise exploratory p-values use Holm correction within the frozen families. The earlier PhysioNet primary contrast uses 20,000 draws with seed 20260924 and an exact two-sided binomial sign test excluding exact zero differences. Its primary hypothesis is shared-minus-broad; contextual comparisons with CSP do not replace it. A supplementary numerical audit uses 200,000 draws with seed 20261002 and is labelled separately. An interval excluding zero need not agree with every small-sample test. For Q15, the predeclared family contains exactly two shared-minus-broad contrasts, one for each external cohort. Percentile person-bootstrap intervals use 20,000 draws with seed 20260924 [24]. Two-sided person-level sign-flip tests use 20,000 Monte Carlo draws with seed 20261003, assuming sign exchangeability of paired person effects under the null, evaluating the absolute mean, and applying the (exceedances + 1)/(20,000 + 1) correction [25]. Holm adjustment controls this two-contrast family [26]; it does not retroactively include the earlier PhysioNet or internal hypotheses. The smallest attainable Monte Carlo p-value is 1/20,001. The intervals are unadjusted 95% intervals, and CSP comparisons are descriptive context. Descriptive Q15 counts of positive, negative, and tied person effects use an absolute 10^−15 tolerance for ties; the frozen primary bootstrap and sign-flip calculations retain the saved numerical effects.~~

For each archived experiment, we report its original statistics and the interval resampling procedure. Rank-selection intervals use 20,000 person-bootstrap draws with seed 20260923. The archived shared-minus-broad representation contrast and robustness intervals use 20,000 draws with seed 20260924. Source-composition and matched-runtime intervals use 10,000 draws with seed

20260926 and exhaustive two-sided sign flips over nine participant differences; familywise exploratory p-values use Holm correction within the frozen families. The earlier PhysioNet primary contrast uses 20,000 draws with seed 20260924 and an exact two-sided binomial sign test excluding exact zero differences. Its primary hypothesis is shared-minus-broad; contextual comparisons with CSP do not replace it. A supplementary numerical audit uses 200,000 draws with seed 20261002 and is labelled separately. An interval excluding zero need not agree with every small-sample test. For the Cho2017/Lee2019 evaluation, the predeclared family contains exactly two shared-minus-broad contrasts, one for each external cohort. Percentile person-bootstrap intervals use 20,000 draws with seed 20260924 [32]. Two-sided person-level sign-flip tests use 20,000 Monte Carlo draws with seed 20261003, assuming sign exchangeability of paired person effects under the null, evaluating the absolute mean, and applying the (exceedances + 1)/(20,000 + 1) correction [33]. Holm adjustment controls this two-contrast family [34]; it does not retroactively include the earlier PhysioNet or internal hypotheses. The smallest attainable Monte Carlo p-value is 1/20,001. The intervals are unadjusted 95% intervals, and CSP comparisons are descriptive context. Descriptive Cho2017/Lee2019 transfer counts of positive, negative, and tied person effects use an absolute 10^−15 tolerance for ties; the frozen primary bootstrap and sign-flip calculations retain the saved numerical effects.

~~The physiological characterization was specified after decoder outcomes were known and remains descriptive. Physiological aggregation and decoder associations use nine person-level records; channels, trials, sessions and seeds are not treated as independent people. The six Spearman coefficients have no physiological p-values or confirmatory testing family. They are separate from Q15's frozen two-cohort tests and cannot retrospectively confirm the decoder contrasts.~~

The physiological characterization was specified after decoder outcomes were known and remains descriptive. Physiological aggregation and decoder associations use nine person-level records; channels, trials, sessions and seeds are not treated as independent people. The six Spearman coefficients have no physiological p-values or confirmatory testing family. They are separate from the Cho2017/Lee2019 evaluation's frozen two-cohort tests and cannot retrospectively confirm the decoder contrasts.

Archived-output CSP diagnostic audit

The diagnostic audit reads the existing frozen CSP prediction tables, probability columns, participant identities and validation hash bindings. It reconstructs participant and pooled confusion matrices without loading a model or producing predictions. Left-prediction proportions and probability medians pool trials; the primary balanced-accuracy estimand remains an equal-person mean. A binary predicted-label complement is an arithmetic check, not a revised predictor or an outcome-selected correction. These post hoc diagnostics do not change primary scores, mappings, thresholds, models or inference contracts. They cannot by themselves exclude every adapter error or identify a physical transfer mechanism.

~~Public BNCI2014_001, PhysioNet, Cho2017 and Lee2019 recordings are available from the original providers [1,2,5,6,7,8,9,10], subject to their licenses and terms of use. PhysioNet platform publications are cited separately [27,28]. Analysis code, saved predictions, participant-level summaries, parameter specifications and numerical validation reports are available at https://github.com/jackzhu119/cross-subject-mi-eeg. The supplementary reproducibility documentation identifies the exact source and result revisions. Raw EEG is not redistributed in the manuscript bundle.~~

Public BNCI2014_001, PhysioNet, Cho2017 and Lee2019 recordings are available from the original providers [1,2,5,6,7,8,9,10], subject to their licenses and terms of use. PhysioNet platform publications are cited separately [35,36]. Analysis code, saved predictions, participant-level summaries, parameter specifications and numerical validation reports are available at https://github.com/jackzhu119/cross-subject-mi-eeg. The supplementary reproducibility documentation identifies the exact source and result revisions. Raw EEG is not redistributed in the manuscript bundle.

~~The current submission materials and licensing notices are available at https://github.com/jackzhu119/cross-subject-mi-eeg/tree/plos-one-submission-v1.1. Author-owned original software is licensed under MIT; author-owned original manuscript material and derived outputs are licensed under CC BY 4.0. Third-party recordings, software, fonts and other incorporated materials retain their original terms. These permissions do not alter frozen scientific results or grant third-party rights.~~

The preserved submitted materials and licensing notices are available at https://github.com/jackzhu119/cross-subject-mi-eeg/tree/plos-one-submission-v1.1. This post-submission revision candidate is maintained separately on paper/plos-one-postsubmission-review-20261010. Author-owned original software is licensed under MIT; author-owned original manuscript material and derived outputs are licensed under CC BY 4.0. Third-party recordings, software, fonts and incorporated materials retain their original terms. Raw EEG is not redistributed.

~~The underlying Codex model version is not reliably documented in the project records. Saved-result checks compared reported numbers with frozen outputs and archived independent validation reports; these checks did not rerun models or physiology experiments during manuscript preparation. The author reports personal review and revision of the scientific content. No AI tool is an author.~~

The underlying Codex model version is not reliably documented in the project records. Manuscript-preparation checks compare saved outputs and archived independent validation reports. The archived-output diagnostic audit involved no model fitting, checkpoint inference or new physiological experiment. The author reports personal verification and revision of the scientific content. No AI tool is an author.

~~The corresponding saved summary is presented in Table 4.~~

Table 4 summarizes completed development conditions.

~~The corresponding saved summary is presented in Fig 2.~~

Fig 2 shows source-selected schedules, participant scores and the one-person duration diagnostic.

~~The corresponding saved summary is presented in Fig 3.~~

Fig 3 displays source-count trajectories and matched-runtime participant differences.

~~The internal shared mu/beta condition achieved 42.19% balanced accuracy, compared with 42.67% for the broad mean-rank baseline, a −0.48 pp mean difference. The archived Q10-V001 exploratory paired interval is [−3.21,+1.41] pp (20,000 draws, seed 20260924), with exact sign-flip p = 0.914. This comparison did not demonstrate a spectral-sharing gain. The restricted broad band and~~

~~individual bands provide descriptive frequency controls, while fixed-duration representation controls retain their declared schedules.~~

The internal shared mu/beta condition achieved 42.19% balanced accuracy, compared with 42.67% for the broad mean-rank baseline, a −0.48 pp mean difference. The archived exploratory paired interval is [−3.21,+1.41] pp (20,000 draws, seed 20260924), with exact sign-flip p = 0.914. This comparison did not demonstrate a spectral-sharing gain. The restricted broad band and individual bands provide descriptive frequency controls, while fixed-duration representation controls retain their declared schedules.

~~The corresponding saved summary is presented in Table 5.~~

Table 5 retains adverse and null robustness results.

~~| Whitening | 44.34 | Q8 historical | +1.67 | [−0.39, +3.67] | 0.3594 |~~

| Whitening | 44.34 | Historical mean-rank | +1.67 | [−0.39, +3.67] | 0.3594 |

~~| Balanced ERM | 43.18 | Q8 historical | +0.51 | [−0.71, +1.77] | 0.4766 |~~

| Balanced ERM | 43.18 | Historical mean-rank | +0.51 | [−0.71, +1.77] | 0.4766 |

~~| Channel dropout | 41.00 | Q8 historical | −1.67 | [−4.51, +0.64] | 0.7617 |~~

| Channel dropout | 41.00 | Historical mean-rank | −1.67 | [−4.51, +0.64] | 0.7617 |

~~| Gain perturbation | 43.42 | Q8 historical | +0.75 | [−0.34, +1.86] | 0.7617 |~~

| Gain perturbation | 43.42 | Historical mean-rank | +0.75 | [−0.34, +1.86] | 0.7617 |

~~| Dropout + gain | 40.92 | Q8 historical | −1.75 | [−5.05, +0.91] | 0.7617 |~~

| Dropout + gain | 40.92 | Historical mean-rank | −1.75 | [−5.05, +0.91] | 0.7617 |

~~Intervals are the archived, unadjusted person-bootstrap intervals; p-values use the declared within-family Holm correction. Every comparison with Q8 crosses runtimes. GroupDRO versus balanced ERM matches batching within the same runtime. Neither intervals nor p-values are used to select a model on target outcomes.~~

Intervals are the archived, unadjusted person-bootstrap intervals; p-values use the declared within-family Holm correction. Every comparison with mean-rank follow-up crosses runtimes. GroupDRO versus balanced ERM matches batching within the same runtime. Neither intervals nor p-values are used to select a model on target outcomes.

~~The corresponding saved summary is presented in Fig 4.~~

Table 6 reports the separate PhysioNet model estimates.

Table 6. Separate binary external evaluation (109 participants).

| Frozen model | Mean BA (%) | Subject SD (pp) | Median BA (%) |

| --- | --- | --- | --- |

| Broad EEGNet | 61.81 | 12.56 | 58.23 |

| Shared mu/beta | 62.39 | 12.92 | 58.73 |

| CSP4 + LDA | 54.46 | 7.90 | 52.08 |

Deep-model scores are averaged over seeds within person. The primary shared-minus-broad contrast is +0.573 pp, 95% CI [−0.097,+1.242], p = 0.06446. CSP provides context and is outside the primary hypothesis. The Methods describe the sampling-rate amendment and versioned portability validation.

Completed Cho2017 and Lee2019 transfer: lower balanced accuracy on Cho2017 and an uncertain Lee2019 contrast

The completed Cho2017/Lee2019 outputs cover 21,320 unique external trials and 149,240 saved model/seed prediction rows, seven per trial. Independent checks verified canonical labels and event identities, person-level confusion matrices, aggregate balanced accuracies, frozen statistical outputs, and links to the declared artifacts. Original source fits remained 15, with zero new migration fits and zero target fits. These counts confirm source-only fitting; they do not constitute independent repetitions of the external experiment.

On Cho2017, equal-person balanced accuracy was 59.82% for broad EEGNet and 58.31% for shared mu/beta EEGNet. The primary shared-minus-broad mean was −1.513 pp (95% person-bootstrap interval [−2.249,−0.804] pp), with two-sided Monte Carlo sign-flip p = 0.000049998 and two-cohort Holm-adjusted p = 0.000099995. The raw p-value is the Monte Carlo procedure's minimum attainable value and does not resolve probabilities below that value. Twelve people favoured shared, 37 favoured broad, and three were tied within a 10^−15 numerical tolerance. Shared performed worse in this declared complete-pipeline comparison after correction.

On Lee2019, broad and shared models achieved 65.51% and 65.72% balanced accuracy. The primary difference was +0.204 pp (95% interval [−0.515,+0.969] pp); raw and Holm-adjusted p-values were both 0.60367. Twenty-three people favoured shared, 30 favoured broad, and one tied. This small, uncertain contrast establishes neither a reliable improvement nor equivalence. The two sessions were combined within each person, so neither the 108 session records nor repeated model seeds increase the primary sample size beyond 54.

Table 7 shows cohort scores and class recalls.

Table 7. Frozen external binary evaluation on Cho2017 and Lee2019.

| Cohort / model | BA (%) | Person SD (pp) | Left recall (%) | Right recall (%) |

| --- | --- | --- | --- | --- |

| Cho2017 / Broad EEGNet | 59.82 | 8.64 | 46.79 | 72.86 |

| Cho2017 / Shared mu/beta | 58.31 | 7.22 | 37.52 | 79.10 |

| Cho2017 / CSP4 + LDA | 51.99 | 5.67 | 94.10 | 9.88 |

| Lee2019 / Broad EEGNet | 65.51 | 10.97 | 75.56 | 55.46 |

| Lee2019 / Shared mu/beta | 65.72 | 10.76 | 72.69 | 58.75 |

| Lee2019 / CSP4 + LDA | 52.77 | 5.57 | 85.98 | 19.56 |

Neural scores and class recalls are averaged over three seeds within person, then equally over people. Person SD describes heterogeneity, not a standard error. Lee includes only offline-training runs, pooled across two sessions. The primary shared-minus-broad contrasts are −1.513 pp [−2.249,−0.804], Holm p = 0.000100 for Cho, and +0.204 pp [−0.515,+0.969], Holm p = 0.604 for Lee. Class recalls and CSP comparisons are descriptive; no additional confirmatory tests were added.

The CSP comparator was near binary chance on both cohorts and strongly favoured left-hand labels. On Cho2017, mean left-hand recall fell from 46.79% with broad input to 37.52% with shared input, while right-hand recall increased from 72.86% to 79.10%. On Lee2019, the corresponding recalls changed from 75.56% to 72.69% and from 55.46% to 58.75%. Balanced accuracy summarizes these asymmetric shifts but does not show which class gains or loses recall. These descriptive checks used the frozen outputs and did not trigger any threshold, label, adapter, or model change. A post-submission audit of the unchanged prediction files found trial-pooled left-prediction proportions of 92.07% for Cho2017 and 83.21% for Lee2019. Median stored left-class probabilities were 0.669 and 0.633. These probability summaries do not assess calibration. Cho participants have unequal trial counts, so pooled prediction proportions must not be confused with the primary equal-person balanced accuracy. Participant confusion matrices and aggregation definitions are provided in S1 Appendix, Table S5 and S1 Data.

Fig 5 presents the frozen Cho2017/Lee2019 complete-pipeline contrasts.

Fig 5. Frozen Cho2017/Lee2019 transfer external evaluation. Cohorts are shown separately: person/model scores, equal-person means and person standard deviations, and frozen paired person-bootstrap 95% intervals. Deep-model scores average three seeds within person; Lee sessions are pooled before person scoring. Holm p-values refer to the predeclared two-cohort shared-minus-broad family. The common 21-channel, two-second pipeline differs from the earlier PhysioNet evaluation, and no pooled cross-cohort test was specified.

Descriptive BNCI Sensor Physiology and Decoder Associations

All 18 raw files, 108 labelled runs and 5,184 trials passed the fixed coverage checks. All 228,096 trial-channel-band rows had finite, positive baseline and task power. No trial, session or person was excluded; the 488 provider-flagged trials remained in the primary summaries. Fig 6 shows equal-person scalp means and all nine C3/Cz/C4 profiles. Task/baseline changes calculated with the fixed estimator varied by sensor, imagined hand and person.

Fig 6. Descriptive native BNCI task/baseline power changes. A–D: equal-person hand-imagery means for mu (8–13 Hz) and beta (13–30 Hz) at all 22 EEG sensors. In the schematic standard-1020 geometry, anterior is upward and left scalp is on the viewer's left; linear interpolation is restricted to the sensor convex hull. No sensor or interpolated grid value is clipped by the fixed ±6-dB range. E–F: all nine equal-session C3/Cz/C4 profiles (thin lines) and equal-person means (thick lines). Trial dB uses cue baseline [−1.5,−0.5) s and task [0.5,2.5) s, with one versus three overlapping Welch segments. Negative values indicate lower task power under this estimator; unequal window precision can bias log ratios. All trials and artifact flags are retained. Template topography neither localizes cortical sources nor attributes decoder decisions. The fixed ±6-dB scale makes the small magnitude of the cohort means visible without rescaling each panel. Circle/triangle symbols

distinguish imagined hands in the profiles; signed values and the shared labelled scale supplement colour. This schematic sensor map is not a cortical source map.

Mu signed laterality averaged −0.250 dB, with six negative and three positive person descriptors; beta averaged −0.168 dB, with eight negative and one positive descriptor (Table 8). Contralateral and ipsilateral group means were both negative in each band, but absolute contralateral changes were not negative for every person. For example, S2 had negative mu laterality (−0.203 dB) even though both its contralateral (+1.491 dB) and ipsilateral (+1.694 dB) means were positive. Separate session, hand and flagged/unflagged summaries are available in the verified supplementary tables as descriptive sensitivity checks; they do not replace the all-trial population.

Table 8. Baseline-relative hand power and signed C3/C4 laterality.

| Band | Contra. mean (dB) | Ipsi. mean (dB) | Mean L (dB) | Person L range (dB) | L signs (−/+) |

| --- | --- | --- | --- | --- | --- |

| Mu, 8–13 Hz | −0.308 | −0.058 | −0.250 | −1.432 to +0.336 | 6 / 3 |

| Beta, 13–30 Hz | −0.439 | −0.272 | −0.168 | −0.519 to +0.133 | 8 / 1 |

Each of nine people receives equal weight, using equal-session mean trial dB from both hands. Contra. and ipsi. average the corresponding C3/C4 hand components; L is contra. minus ipsi. All 2,592 hand trials, including 246 flagged trials, are retained. Negative L means a lower contralateral than ipsilateral task/baseline change; it does not establish an absolute contralateral power decrease. Values are rounded to three decimals here and retained at full precision in the data tables. These fixed-estimator ratios are descriptive, not unbiased ERD estimates.

For the designated broad-EEGNet endpoint, descriptive associations between signed laterality and saved binary LOSO balanced accuracy were rho = −0.150 for mu and +0.600 for beta. Shared-input coefficients were −0.083 and +0.483, and CSP4+LDA coefficients were +0.067 and +0.317, respectively (Table S4; S1 Appendix, Fig S3). All six coefficients use the same nine people and have no p-values. The positive beta coefficient indicates that higher balanced accuracy accompanied a larger, less-negative signed descriptor; it does not support a benefit from stronger relative contralateral suppression.

Discussion

Training-duration Sensitivity in a Small Development Benchmark

Training duration was a major source of internal performance variation. At two epochs, S3 remained near chance with either raw or source-normalized inputs; at 16 epochs, both performed substantially better. Rank aggregation changed the source-selected schedule and recovered performance for S3 and S8. The matched-runtime control also retained a large mean difference between fixed training and the historical mean-loss schedule. An apparent cross-person model collapse should therefore be examined alongside selected duration, source-validation trajectories, per-class recall and prediction concentration. These outputs help distinguish an inadequate stopping schedule from a failure attributed prematurely to the input representation. In this benchmark, a fixed 20-epoch development condition also reached a cohort mean of 43.99%, compared with 42.67% for mean-rank and 33.72% for the earlier mean-loss condition. Those cross-batch means are descriptive. The matched-runtime

evidence shows that duration auditing can diagnose apparent collapse without establishing an optimal stopping rule or a typical-person benefit.

These findings do not establish rank selection as a general solution. Within-fold ranks remove loss-scale differences but discard the size of those differences; equal fold weighting imposes a further aggregation choice. The candidate was developed after this benchmark had been inspected. In addition, the matched-runtime fixed-duration condition improved the mean while the median paired difference was negative. The external evaluations did not compare rank with mean-loss selection and cannot independently confirm the internal gain. A prospective test would fix the competing selection rules before observing new-person outcomes and match runtime, source examples and update budgets.

Complete Spectral Pipelines Have Cohort-dependent Outcomes

The frozen evaluations limit claims of a consistent shared-input advantage. Shared mu/beta input gave no internal advantage, a small uncertain positive estimate on PhysioNet, a corrected adverse contrast on Cho2017 and an uncertain near-zero contrast on Lee2019. The Cho2017 finding contradicts a consistent advantage of this fixed shared-input pipeline under the evaluated source-only adapter. It does not show that mu/beta rhythms lack information, that parameter sharing is generally harmful or that a different band definition would perform better.

The class recalls reveal a trade-off hidden by the cohort means. On Cho2017, the shared-input pipeline reduced mean left-hand recall while increasing right-hand recall; on Lee2019, opposing recall changes partly offset one another. These patterns describe the frozen predictions. They do not justify correcting labels after inspection or attributing the contrast to physiology. Montage, acquisition reference, amplitude scale, temporal alignment and band-logit averaging are possible contributors, but the study did not isolate any of them.

One shared parameter set does not imply a cheaper computation: the network processes two band views and pools their BatchNorm exposure. Independent-branch controls alter both capacity and normalization, and the broadband capacity comparator retains a small disclosed parameter mismatch. The source-fitted CSP model also transfers poorly and asymmetrically in the Cho2017/Lee2019 evaluation. Performance above this contextual comparator does not establish practical or clinical reliability. Contemporary adaptation and foundation-model methods were not compared under a common budget, so the results support no state-of-the-art claim.

The external schedules belong to the complete-pipeline contrast: broad/shared fits used 18/17 epochs for PhysioNet and 14/19 for Cho2017/Lee2019. The paired differences combine duration selection, preprocessing, one-view versus two-view computation and normalization exposure. They cannot identify the independent effect of parameter sharing or band decomposition. No matched-duration pair of final all-source checkpoints was available in the frozen evidence; comparing a new equal-duration pair would require additional source fits if those models do not already exist.

The archived-output audit confirms a severe CSP class asymmetry, not an exact single-class prediction for every trial. Original source validation establishes artifact and partition consistency, and the external replay establishes consistency of the declared adapter and frozen predictions. Neither proves that the chosen CSP design is optimal or that every acquisition and mapping assumption is physically correct. Held-out source predictions for this exact 21-channel CSP specification were not available in the audited evidence. Transfer sensitivity is a plausible

explanation, but neither an exclusive causal account nor the absence of every implementation error is established.

Person-level Evidence and Descriptive Signal Characterization

Additional seeds assess sensitivity to stochastic fitting; they do not increase the number of participants. Lee2019 sessions and source-identity subsets likewise provide repeated observations within a person. The paired person-level analysis retains all declared participants and reports the direction of each difference. Its bootstrap intervals condition on the frozen source models and omit uncertainty from another nine-person source cohort, acquisition adapter or development history. For the internal study, overlapping outer training sets further limit an independent-person interpretation.

A few large recoveries can improve a mean without producing consistent benefit across people, so tests of average change and directional consistency need not agree. The original rank-selection sign calculation retained exact ties; excluding them conventionally gives a different result from both the archived calculation and the paired t-test. The discrepancy must remain visible rather than be hidden under an incorrect test label. The later internal families retain their corrected null findings. Percentile bootstrap intervals and randomization tests also need not agree in a small sample; an interval excluding zero does not override a corrected null test. The Cho2017/Lee2019 evaluation uses a separately frozen two-cohort family: Cho2017 survives correction, whereas Lee2019 establishes neither benefit nor equivalence. These separate tests do not establish a cohort-by-pipeline interaction. PhysioNet has a different archived primary test and an amended protocol; its p-value belongs to that evaluation rather than the newer family.

The BNCI analysis provides descriptive, sensor-level baseline-relative characterization, not the cause of the decoder contrasts. Mean signed hand laterality was negative in both bands: six of nine people had negative mu laterality and eight had negative beta laterality, with positive signs in the others. A negative relative descriptor can still accompany increased power on both sides, as in S2, so its absolute components must also be examined. The positive broad-model beta correlation means higher balanced accuracy accompanied less-negative relative laterality, rather than stronger suppression. None of the six associations supports a subgroup threshold or a model-selection rule.

Neither a lateralized scalp mean nor its correlation with LOSO balanced accuracy locates a cortical source, excludes ocular or muscle contributions, or demonstrates that EEGNet used that signal. This BNCI-only physiology analysis was specified after decoder outcomes were known. It cannot explain the adverse Cho contrast or validate an external physiological mechanism.

Headline accuracies from other studies cannot position these cohort means in a common ranking. Roots et al. report pooled-window accuracy rather than participant-level balanced accuracy; overlapping-filter and multiscale architectures use different branches, tasks and selection procedures [23,24,28]. Target fine-tuning and calibration further distinguish adaptive transfer and BEETL [26,27]. Cohort eligibility, trial window, reference, montage, training exposure and target-information access must all match before a numerical comparison can address method superiority. No published model was rerun under such a matched budget.

Limitations

Successive development conditions used the same nine BNCI people. Excluding the target from each fit does not remove outcome-informed candidate design or dependence through overlapping LOSO

training sets. The internal rank-selection gain was concentrated in two people, and the external evaluations did not test that selection-rule contrast. Controls also differ in runtime, training examples, update exposure, sensor width or approximate capacity. There is no common-budget comparison with contemporary adaptation or foundation-model methods.

The PhysioNet and Cho2017/Lee2019 stages differ in montages, windows, references, filtering scope and source checkpoints. The earlier PhysioNet evaluation's metadata amendment followed partial execution. In the frozen transfer experiment, repository freezes preceded fitting or inference but followed metadata inspection; they are not independently registered preregistration. Cross-provider participant identities remain unverified. The external bootstrap intervals condition on one frozen source cohort and omit source-sampling, adapter and development-history uncertainty.

Fig 4 reports the earlier PhysioNet evaluation separately.

~~The corresponding saved summary is presented in Table 6.~~

~~**Table 6. Separate binary external evaluation (109 participants).**~~

~~| Frozen model | Mean BA (%) | Subject SD (pp) | Median BA (%) |~~

~~| --- | --- | --- | --- |~~

~~| Broad EEGNet | 61.81 | 12.56 | 58.23 |~~

~~| Shared mu/beta | 62.39 | 12.92 | 58.73 |~~

~~| CSP4 + LDA | 54.46 | 7.90 | 52.08 |~~

~~Deep-model scores are averaged over seeds within person. The primary shared-minus-broad contrast is +0.573 pp, 95% CI [−0.097,+1.242], p = 0.06446. CSP provides context and is outside the primary hypothesis. The Methods describe the sampling-rate amendment and versioned portability validation.~~

~~### Completed Cho2017 and Lee2019 transfer: lower balanced accuracy on Cho2017 and an uncertain Lee2019 contrast~~

~~The completed Q15 outputs cover 21,320 unique external trials and 149,240 saved model/seed prediction rows, seven per trial. Independent checks verified canonical labels and event identities, person-level confusion matrices, aggregate balanced accuracies, frozen statistical outputs, and links to the declared artifacts. Original source fits remained 15, with zero new migration fits and zero target fits. These counts confirm source-only fitting; they do not constitute independent repetitions of the external experiment.~~

~~On Cho2017, equal-person balanced accuracy was 59.82% for broad EEGNet and 58.31% for shared mu/beta EEGNet. The primary shared-minus-broad mean was −1.513 pp (95% person-bootstrap interval [−2.249, −0.804] pp), with two-sided Monte Carlo sign-flip p = 0.000049998 and two-cohort Holm-adjusted p = 0.000099995. The raw p-value is the Monte Carlo procedure's minimum attainable value and does not resolve probabilities below that value. Twelve people favoured shared, 37 favoured broad, and three were tied within a 10^−15 numerical tolerance. Shared performed worse in this declared complete-pipeline comparison after correction.~~

On Lee2019, broad and shared models achieved 65.51% and 65.72% balanced accuracy. The primary difference was +0.204 pp (95% interval [−0.515,+0.969] pp); raw and Holm-adjusted p-values were both 0.60367. Twenty-three people favoured shared, 30 favoured broad, and one tied. This small, uncertain contrast establishes neither a reliable improvement nor equivalence. The two sessions were combined within each person, so neither the 108 session records nor repeated model seeds increase the primary sample size beyond 54.

The corresponding saved summary is presented in Table 7.

Table 7. New external binary evaluation under the Q15 freeze.

| Cohort / model | BA (%) | Person SD (pp) | Left recall (%) | Right recall (%) |

|---|---|---|---|---|

| Cho2017 / Broad EEGNet | 59.82 | 8.64 | 46.79 | 72.86 |

| Cho2017 / Shared mu/beta | 58.31 | 7.22 | 37.52 | 79.10 |

| Cho2017 / CSP4 + LDA | 51.99 | 5.67 | 94.10 | 9.88 |

| Lee2019 / Broad EEGNet | 65.51 | 10.97 | 75.56 | 55.46 |

| Lee2019 / Shared mu/beta | 65.72 | 10.76 | 72.69 | 58.75 |

| Lee2019 / CSP4 + LDA | 52.77 | 5.57 | 85.98 | 19.56 |

Neural scores and class recalls are averaged over three seeds within person, then equally over people. Person SD describes heterogeneity, not a standard error. Lee includes only offline-training runs, pooled across two sessions. The primary shared-minus-broad contrasts are −1.513 pp [−2.249,−0.804], Holm p = 0.000100 for Cho, and +0.204 pp [−0.515,+0.969], Holm p = 0.604 for Lee. Class recalls and CSP comparisons are descriptive; no additional confirmatory tests were added.

The CSP comparator was near binary chance on both cohorts and strongly favoured left-hand labels. On Cho2017, mean left-hand recall fell from 46.79% with broad input to 37.52% with shared input, while right-hand recall increased from 72.86% to 79.10%. On Lee2019, the corresponding recalls changed from 75.56% to 72.69% and from 55.46% to 58.75%. Balanced accuracy summarizes these asymmetric shifts but does not show which class gains or loses recall. These descriptive checks used the frozen outputs and did not trigger any threshold, label, adapter, or model change.

The corresponding saved summary is presented in Fig 5.

Fig 5. Frozen external evaluation on Cho2017 and Lee2019. Frozen Q15 external evaluation. Cohorts are shown separately: person/model scores, equal-person means and person standard deviations, and frozen paired person-bootstrap 95% intervals. Deep-model scores average three seeds within person; Lee sessions are pooled before person scoring. Holm p-values refer to the predeclared two-cohort shared-minus-broad family. The common 21-channel, two-second pipeline differs from Q14, and no pooled cross-cohort test was specified.

Descriptive BNCI Sensor-Physiology and Decoder Associations (Q16)

All 18 raw files, 108 labelled runs and 5,184 trials passed the fixed coverage checks. All 228,096 trial-channel-band rows had finite, positive baseline and task power. No trial, session or person was excluded; the 488 provider-flagged trials remained in the primary summaries. Fig 6 shows equal-person scalp means and all nine C3/Cz/C4 profiles. Task/baseline changes calculated with the fixed estimator varied by sensor, imagined hand and person.

Fig 6. Descriptive native BNCI task/baseline power changes. A–D: equal-person hand-imagery means for mu (8–13 Hz) and beta (13–30 Hz) at all 22 EEG sensors. In the schematic standard-1020 geometry, anterior is upward and left scalp is on the viewer's left; linear interpolation is restricted to the sensor convex hull. No sensor or interpolated grid value is clipped by the fixed ±6 dB range. E–F: all nine equal-session C3/Cz/C4 profiles (thin lines) and equal-person means (thick lines). Trial dB uses cue baseline [−1.5, −0.5) s and task [0.5,2.5) s, with one versus three overlapping Welch segments. Negative values indicate lower task power under this estimator; unequal window precision can bias log ratios. All trials and artifact flags are retained. Template topography neither localizes cortical sources nor attributes decoder decisions.

Mu signed laterality averaged −0.250 dB, with six negative and three positive person descriptors; beta averaged −0.168 dB, with eight negative and one positive descriptor (Table 8). Contralateral and ipsilateral group means were both negative in each band, but absolute contralateral changes were not negative for every person. For example, S2 had negative mu laterality (−0.203 dB) even though both its contralateral (+1.491 dB) and ipsilateral (+1.694 dB) means were positive. Separate session, hand and flagged/unflagged summaries are available in the verified supplementary tables as descriptive sensitivity checks; they do not replace the all-trial population.

Table 8. Baseline-relative hand power and signed C3/C4 laterality.

| Band | Contra. mean (dB) | Ipsi. mean (dB) | Mean L (dB) | Person L range (dB) | L signs (−/+) |

|---|---|---|---|---|---|

| Mu, 8–13 Hz | −0.308 | −0.058 | −0.250 | −1.432 to +0.336 | 6 / 3 |

| Beta, 13–30 Hz | −0.439 | −0.272 | −0.168 | −0.519 to +0.133 | 8 / 1 |

Each of nine people receives equal weight, using equal-session mean trial dB from both hands. Contra. and ipsi. average the corresponding C3/C4 hand components; L is contra. minus ipsi. All 2,592 hand trials, including 246 flagged trials, are retained. Negative L means a lower contralateral than ipsilateral task/baseline change; it does not establish an absolute contralateral power decrease. Values are rounded to three decimals here and retained at full precision in the data tables. These fixed-estimator ratios are descriptive, not unbiased ERD estimates.

For the designated broad-EEGNet endpoint, descriptive associations between signed laterality and saved binary LOSO balanced accuracy were rho = −0.150 for mu and +0.600 for beta. Shared-input coefficients were −0.083 and +0.483, and CSP4+LDA coefficients were +0.067 and +0.317, respectively (Table S4; S1 Appendix, Fig S3). All six coefficients use the same nine people and have no p-values. The positive beta coefficient indicates that higher balanced accuracy accompanied a larger, less-negative signed descriptor; it does not support a benefit from stronger relative contralateral suppression.

Discussion

Training-duration Sensitivity in a Small Development Benchmark

Training duration was a major source of internal performance variation. At two epochs, S3 remained near chance with either raw or source-normalized inputs; at 16 epochs, both performed substantially better. Rank aggregation changed the source-selected schedule and recovered performance for S3 and S8. The matched-runtime control also retained a large mean difference between fixed training and the historical mean-loss schedule. An apparent cross-person model collapse should therefore be examined alongside selected duration, source-validation trajectories, per-class recall and prediction concentration. These outputs help distinguish an inadequate stopping schedule from a failure attributed prematurely to the input representation.

These findings do not establish rank selection as a general solution. Within-fold ranks remove loss-scale differences but discard the size of those differences; equal fold weighting imposes a further aggregation choice. The candidate was developed after this benchmark had been inspected. In addition, the matched-runtime fixed-duration condition improved the mean while the median paired difference was negative. The external evaluations did not compare rank with mean-loss selection and cannot independently confirm the internal gain. A prospective test would fix the competing selection rules before observing new-person outcomes and match runtime, source examples and update budgets.

Complete Spectral Pipelines Have Cohort-dependent Outcomes

The frozen evaluations limit claims of a consistent shared-input advantage. Shared mu/beta input gave no internal advantage, a small uncertain positive estimate on PhysioNet, a corrected adverse contrast on Cho2017 and an uncertain near-zero contrast on Lee2019. The Cho2017 finding contradicts a consistent advantage of this fixed shared-input pipeline under the evaluated source-only adapter. It does not show that mu/beta rhythms lack information, that parameter sharing is generally harmful or that a different band definition would perform better.

The class recalls reveal a trade-off hidden by the cohort means. On Cho2017, the shared-input pipeline reduced mean left-hand recall while increasing right-hand recall; on Lee2019, opposing recall changes partly offset one another. These patterns describe the frozen predictions. They do not justify correcting labels after inspection or attributing the contrast to physiology. Montage, acquisition reference, amplitude scale, temporal alignment and band-logit averaging are possible contributors, but the study did not isolate any of them.

One shared parameter set does not imply a cheaper computation: the network processes two band views and pools their BatchNorm exposure. Independent-branch controls alter both capacity and normalization, and the broadband capacity comparator retains a small disclosed parameter mismatch. The source-fitted CSP model also transfers poorly and asymmetrically in Q15. Performance above this contextual comparator does not establish practical or clinical reliability. Contemporary adaptation and foundation-model methods were not compared under a common budget, so the results support no state-of-the-art claim.

The external schedules are part of the pipeline contrast: broad/shared fits used 18/17 epochs in Q14 and 14/19 in Q15. Paired differences therefore combine the selected training durations, input views and shared normalization. They do not identify the causal contribution of parameter sharing alone.

Person-level Evidence and Descriptive Signal Characterization

Additional seeds assess sensitivity to stochastic fitting; they do not increase the number of participants. Lee2019 sessions and source-identity subsets likewise provide repeated observations within a person. The paired person-level analysis retains all declared participants and reports the direction of each difference. Its bootstrap intervals condition on the frozen source models and omit uncertainty from another nine-person source cohort, acquisition adapter or development history. For the internal study, overlapping outer training sets further limit an independent-person interpretation.

A few large recoveries can improve a mean without producing consistent benefit across people, so tests of average change and directional consistency need not agree. The original rank-selection sign calculation retained exact ties; excluding them conventionally gives a different result from both the archived calculation and the paired t-test. The discrepancy must remain visible rather than be hidden under an incorrect test label. The later internal families retain their corrected null findings. Percentile bootstrap intervals and randomization tests also need not agree in a small sample; an interval excluding zero does not override a corrected null test. Q15 uses a separately frozen two-cohort family: Cho2017 survives correction, whereas Lee2019 establishes neither benefit nor equivalence. These separate tests do not establish a cohort-by-pipeline interaction. PhysioNet has a different archived primary test and an amended protocol; its p-value belongs to that evaluation rather than the newer family.

The BNCI signals describe measured sensor activity, not the cause of the decoder contrasts. Mean-signed hand laterality was negative in both bands: six of nine people had negative mu laterality and eight had negative beta laterality, with positive signs in the others. A negative relative descriptor can still accompany increased power on both sides, as in S2, so its absolute components must also be examined. The positive broad-model beta correlation means higher balanced accuracy accompanied less-negative relative laterality, rather than stronger suppression. None of the six associations supports a subgroup threshold or a model-selection rule.

Neither a lateralized scalp mean nor its correlation with LOSO balanced accuracy locates a cortical source, excludes ocular or muscle contributions, or demonstrates that EEGNet used that signal. This BNCI-only physiology analysis was specified after decoder outcomes were known. It cannot explain the adverse Cho contrast or validate an external physiological mechanism.

Limitations

Successive development conditions used the same nine BNCI people. Excluding the target from each fit does not remove outcome-informed candidate design or dependence through overlapping LOSO training sets. The internal rank-selection gain was concentrated in two people, and the external evaluations did not test that selection-rule contrast. Controls also differ in runtime, training examples, update exposure, sensor width or approximate capacity. There is no common-budget comparison with contemporary adaptation or foundation-model methods.

Q14 and Q15 differ in montages, windows, references, filtering scope and source checkpoints. Q14's metadata amendment followed partial execution. In the frozen transfer experiment, repository freezes preceded fitting or inference but followed metadata inspection; they are not independently registered preregistration. Cross-provider participant identities remain unverified. The external bootstrap intervals condition on one frozen source cohort and omit source-sampling, adapter and development-history uncertainty.

~~Q15 used a common operational tensor and a committed adapter for transfer across three providers. Matching tensor dimensions does not make the physical measurements equivalent. Native export voltage calibration, the original Cho acquisition reference and hardware cue latency were not independently verified. The Cho channel order was mapped from the publication montage because the MATLAB payload has no channel-name strings. Lee event indexing retains a one-native-sample convention discrepancy. Numerical consistency checks cannot resolve these acquisition uncertainties.~~

The Cho2017/Lee2019 study used a common operational tensor and a committed adapter for transfer across three providers. Matching tensor dimensions does not make the physical measurements equivalent. Native export voltage calibration, the original Cho acquisition reference and hardware cue latency were not independently verified. The Cho channel order was mapped from the publication montage because the MATLAB payload has no channel-name strings. Lee event indexing retains a one-native-sample convention discrepancy. Numerical consistency checks cannot resolve these acquisition uncertainties.

~~Q14 and Q15 also differ in the extent of verification. The earlier PhysioNet evaluation required a sampling-rate amendment after partial execution; its later portability check examined saved outputs without replaying inference. Q15 froze preprocessing before source fitting and the inference contract before external prediction, and its completed scientific validator reports independent raw-to-prediction replay. The manuscript checks recalculate saved outputs and verify hash bindings. These provide different levels of reproducibility and should not be treated as interchangeable.~~

The two external stages also differ in the extent of verification. The earlier PhysioNet evaluation required a sampling-rate amendment after partial execution; its later portability check examined saved outputs without replaying inference. The Cho2017/Lee2019 study froze preprocessing before source fitting and the inference contract before external prediction, and its completed scientific validator reports independent raw-to-prediction replay. The manuscript checks recalculate saved outputs and verify hash bindings. These provide different levels of reproducibility and should not be treated as interchangeable.

~~Evaluation was offline and used zero-phase filtering. Online latency, prospective assistive control, clinical populations and patient benefit were not tested. The two-second Q15 crop omits part of each imagery interval and differs from the providers' original benchmarks. A subsequent study could preregister a calibrated adapter and frozen source-only models for another cohort, specifying per-class and per-person outcomes before access. Retuning on the present held-out cohorts would be further development, not independent external confirmation.~~

Evaluation was offline and used zero-phase filtering. Online latency, prospective assistive control, clinical populations and patient benefit were not tested. The two-second Cho2017/Lee2019 transfer crop omits part of each imagery interval and differs from the providers' original benchmarks. A subsequent study could preregister a calibrated adapter and frozen source-only models for another cohort, specifying per-class and per-person outcomes before access. Retuning on the present held-out cohorts would be further development, not independent external confirmation.

~~Training-duration selection was consequential in this nine-person BNCI development benchmark, but its gains were concentrated rather than broadly shared. Fixed shared-input pipelines showed no consistent external advantage: balanced accuracy was 1.513 pp lower than broad EEGNet on Cho2017, and the Lee2019 estimate remained compatible with small effects in either direction.~~

~~Unequal source-selected durations are part of these complete-pipeline comparisons, which do not isolate parameter sharing. The post-outcome BNCI sensor analysis found negative mean signed hand laterality in mu and beta with substantial participant variation; it established neither a decoder mechanism nor external physiological replication. Unknown acquisition calibration and offline evaluation limit deployment conclusions.~~

Training-duration selection was consequential in this nine-person BNCI development benchmark, but its gains were concentrated rather than broadly shared. Complete fixed shared-input pipelines showed no consistent external advantage. The Cho2017 shared-input pipeline had 1.513-pp lower balanced accuracy than broad EEGNet under its 19-versus-14-epoch source-selected schedule; Lee2019 remained compatible with small effects in either direction. These comparisons do not isolate spectral sharing. Archived CSP predictions document strong class asymmetry but do not identify its cause. The post-outcome BNCI sensor analysis found negative mean signed hand laterality with participant variation; it establishes neither a decoder mechanism nor external physiological replication. Acquisition uncertainty and offline evaluation limit deployment conclusions.

23. Roots K, Muhammad Y, Muhammad N. Fusion Convolutional Neural Network for Cross-Subject EEG Motor Imagery Classification. Computers. 2020;9(3):72. doi:10.3390/computers9030072

24. Luo J, Li J, Mao Q, Shi Z, Liu H, Ren X, et al. Overlapping filter bank convolutional neural network for multisubject multicategory motor imagery brain-computer interface. BioData Min. 2023;16(1):19. doi:10.1186/s13040-023-00336-y

25. Smedemark-Margulies N, Wang Y, Koike-Akino T, Liu J, Parsons K, Bicer Y, et al. Improving subject transfer in EEG classification with divergence estimation. J Neural Eng. 2024;21(6):066031. doi:10.1088/1741-2552/ad9777

26. Zhang K, Robinson N, Lee SW, Guan C. Adaptive transfer learning for EEG motor imagery classification with deep Convolutional Neural Network. Neural Netw. 2021;136:1-10. doi:10.1016/j.neunet.2020.12.013

27. Wei X, Faisal AA, Grosse-Wentrup M, Gramfort A, Chevallier S, Jayaram V, et al. 2021 BEETL Competition: Advancing Transfer Learning for Subject Independence and Heterogenous EEG Data Sets. Proc Mach Learn Res. 2022;176:205–219. Available from: https://proceedings.mlr.press/v176/wei22a.html [cited 2026 Oct 10].

28. Cao D, Yu Z, Wang J, Wu Y. SMMTM: Motor imagery EEG decoding algorithm using a hybrid multi-branch separable convolutional self-attention temporal convolutional network. PLOS ONE. 2025;20(10):e0333805. doi:10.1371/journal.pone.0333805

29. Zhao Y, He D, Ren F, Xia Q, Xu L, Xie G, et al. RMETNet: A cross-subject motor imagery EEG signal classification model based on TSLANet and riemannian geometry features. PLOS ONE. 2026;21(4):e0347671. doi:10.1371/journal.pone.0347671

30. Sonntag J, Yu L, Wang X, Schack T. Neurophysiological predictors of deep learning based unilateral upper limb motor imagery classification. Front Hum Neurosci. 2025;19:1617748. doi:10.3389/fnhum.2025.1617748

~~23. Sagawa S, Koh PW, Hashimoto TB, Liang P. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. International Conference~~

~~on Learning Representations. 2020. Available from: https://openreview.net/forum?id=ryxGuJrFvS [cited 2026 Oct 8].~~

31. Sagawa S, Koh PW, Hashimoto TB, Liang P. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. International Conference on Learning Representations. 2020. Available from: https://openreview.net/forum?id=ryxGuJrFvS [cited 2026 Oct 8].

~~24. Efron B. Bootstrap Methods: Another Look at the Jackknife. Ann Stat. 1979;7(1). doi:10.1214/aos/1176344552~~

32. Efron B. Bootstrap Methods: Another Look at the Jackknife. Ann Stat. 1979;7(1). doi:10.1214/aos/1176344552

~~25. Winkler AM, Ridgway GR, Webster MA, Smith SM, Nichols TE. Permutation inference for the general linear model. Neuroimage. 2014;92:381-397. doi:10.1016/j.neuroimage.2014.01.060~~

33. Winkler AM, Ridgway GR, Webster MA, Smith SM, Nichols TE. Permutation inference for the general linear model. Neuroimage. 2014;92:381-397. doi:10.1016/j.neuroimage.2014.01.060

~~26. Holm S. A Simple Sequentially Rejective Multiple Test Procedure. Scand J Stat. 1979;6(2):65–70. Available from: https://www.jstor.org/stable/4615733 [cited 2026 Oct 8].~~

34. Holm S. A Simple Sequentially Rejective Multiple Test Procedure. Scand J Stat. 1979;6(2):65–70. Available from: https://www.jstor.org/stable/4615733 [cited 2026 Oct 8].

~~27. Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals. Circulation. 2000;101(23). doi:10.1161/01.CIR.101.23.e215~~

35. Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals. Circulation. 2000;101(23). doi:10.1161/01.CIR.101.23.e215

~~28. Pollard T, Moody BE, Lehman LwH, Gow BJ, Fernandes C, Xie C, et al. PhysioNet as a global platform for biomedical research. Nat Health. 2026;1(8):792-795. doi:10.1038/s44360-026-00096-z~~

36. Pollard T, Moody BE, Lehman LwH, Gow BJ, Fernandes C, Xie C, et al. PhysioNet as a global platform for biomedical research. Nat Health. 2026;1(8):792-795. doi:10.1038/s44360-026-00096-z

~~S1 Appendix. Supplementary results and detailed reproducible methods.~~

S1 Appendix. Supplementary results, detailed methods and post-submission diagnostic context.

~~S1 Data. Frozen derived data and validation evidence.~~

S1 Data. Frozen derived outputs with separately identified archived-output diagnostic additions.